

Project Module 1: Fairness Audit and Bias Mitigation

German Credit Dataset

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**THE AMERICAN
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Part 1 (Fairness Audit)

Dataset

- **Dataset:** German Credit Dataset
- **Task:** Predict creditworthiness (binary classification)
- **Protected Attributes:** Age (under_25 vs age_25_and_over), Gender (female vs male)
- **Typical Fairness Concerns:** Loan approval disparities across demographic groups

This part focuses on auditing the baseline model for fairness and predictive performance across these sensitive groups.

1. Baseline Predictive Performance (Overall)

Baseline Model: Logistic regression model.

Justification:

- Logistic regression is interpretable, making it suitable for fairness analysis since decision boundaries and feature effects are easier to understand.
- It outputs probabilities, which are necessary for evaluating calibration metrics (ECE, Brier score).
- It is a standard benchmark model in credit scoring tasks, providing a reliable baseline before using more complex models.
- It allows us to assess whether fairness issues arise from the data itself, rather than model complexity.

Overall Results on Test Set:

Metric	Value
Accuracy	0.7050
Precision	0.7755
Recall	0.8143
F1-score	0.7944

Interpretation of metrics:

- **Accuracy = 0.7050** → About **70.5% of all test people** are predicted correctly (either correctly approved or correctly rejected). Accuracy alone can **hide fairness issues** because it averages across groups.
- **Precision = 0.7755** → Among predictions of creditworthy ($\hat{y} = 1$), **77.55% are truly creditworthy**. Approvals are fairly reliable, but not perfect.
- **Recall = 0.8143** → Among truly creditworthy people ($y = 1$), the model identifies **81.43%**. Captures most qualified applicants.
- **F1-score = 0.7944** → The harmonic mean of precision and recall. It provides a **balanced measure** of performance, especially useful when class distribution is uneven.

Overall accuracy is reasonable, but fairness must be checked separately, as different demographic groups may be approved at different rates or misclassified more often.

2. Fairness Table Columns

For each group in the fairness tables:

Metric	Definition	Interpretation
selection_rate	$P(\hat{y} = 1 \mid \text{group})$	This is the approval rate of the model for that group.
tpr	True Positive Rate	among people with $y = 1$ in that group, how many get predicted as $\hat{y} = 1$. (Higher tpr is better for catching real creditworthy cases.)
fpr	False Positive Rate	among people with $y = 0$ in that group, how many get predicted as $\hat{y} = 1$. (Higher fpr means more incorrect approvals.)
ece	Expected Calibration Error	Average difference between predicted probabilities and true outcomes (lower is better)
brier	Mean squared error between predicted probability and true label	Lower values indicate better probability quality

- **Demographic parity** (selection_rate) $\rightarrow$ Who gets approved more
- **Equalized Odds (TPR and FPR)** $\rightarrow$ How accurately the model treats true positives/negatives
- **Calibration (ECE + Brier)** $\rightarrow$ Whether the model's probability outputs are reliable

3. Fairness Audit by Gender

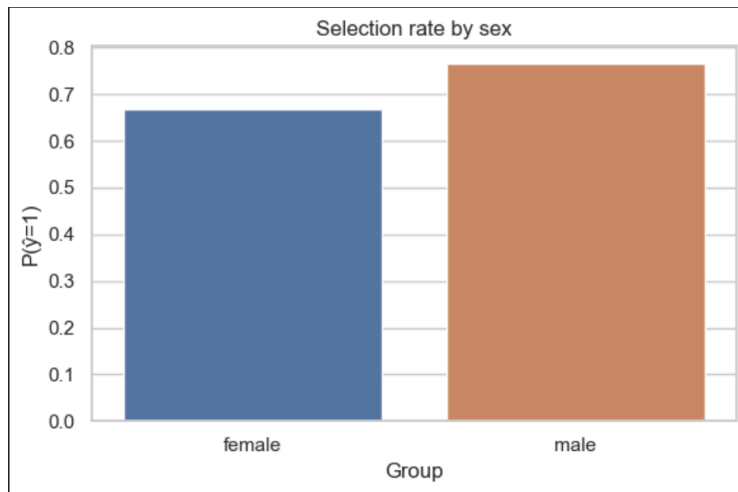
Group Metrics:

Group	Selection Rate	TPR	FPR	ECE	Brier
Female	0.6667	0.725	0.55	0.1809	0.2293
Male	0.7643	0.850	0.55	0.0638	0.1644

3.1 Demographic Parity (Selection Rate)

- Female approval rate = 0.6667
- Male approval rate = 0.7643
- Gap = 0.0976 approximately 0.1 $\rightarrow$ **Demographic parity is met** at the edge of the threshold

Selection Rate Bars Figure (Demographic Parity)



- Higher bars $\rightarrow$ higher approval
- Gender: Male $>$ Female

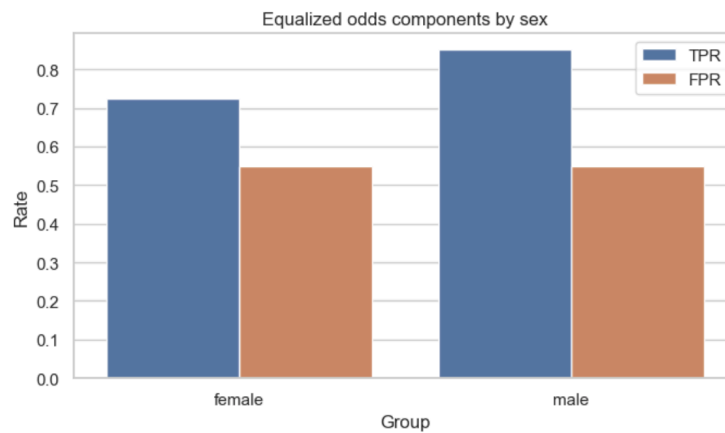
3.2 Equalized Odds (TPR and FPR)

- TPR gap = 0.1250 (male 0.850 vs female 0.725)
- FPR gap = 0.0000 (both 0.55)

Interpretation:

- Males are **better identified as truly creditworthy**
- Incorrect approvals are similar across genders

Violation Rule: TPR or FPR gap = 0.125 $>$ 0.1 $\rightarrow$ **Equalized odds NOT met**
Equalized Odds Bars Plot (TPR & FPR)



- Differences in TPR $\rightarrow$ group detection disparities
- Differences in FPR $\rightarrow$ incorrect approval disparities
- Gender: TPR differs, FPR similar
- Age: Both TPR and FPR differ strongly

3.3 Calibration (ECE + Brier)

Metrics:

- Female ECE = 0.1809, Male ECE = 0.0638 → **Gap = 0.1172 > 0.1**
- Female Brier = 0.2293 > Male Brier = 0.1644 (Less is better)

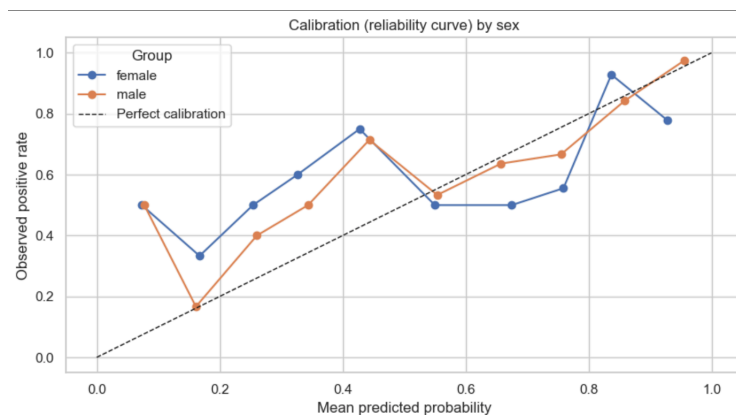
So, calibration **NOT** met

Interpretation:

- **Expected Calibration Error (ECE):** For females, ECE = 0.1809 indicates that the model's probability estimates are **less aligned with reality** compared to males (ECE = 0.0638) with a gap of 0.1172 > 0.1 shows a significant difference in probability reliability across genders.
- **Brier Score:** Higher Brier score for females (0.2293 vs 0.1644) indicates **less accurate probability estimates**

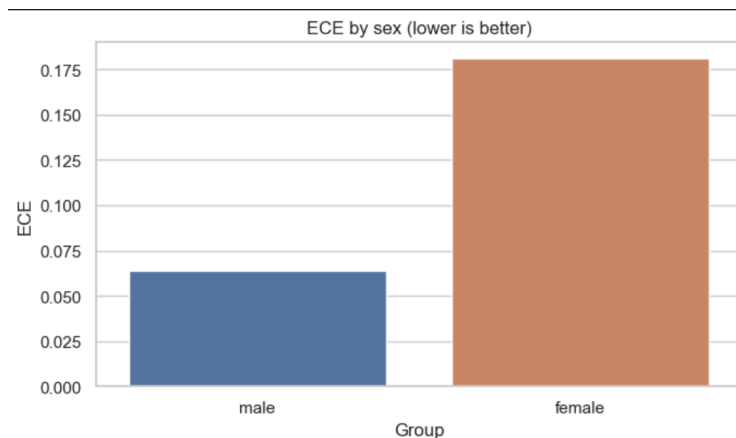
So, the model's probability outputs are **significantly less reliable for females**, which could affect decision-making confidence in real-world loan approvals.

Calibration Plot (Reliability Curves)



- Reliability curves closer to diagonal → better calibration
- Gender: Female worse than male

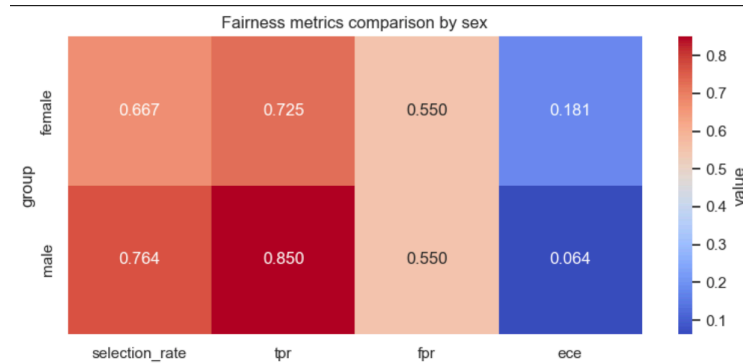
Calibration Plot (ECE)



- Higher ECE → worse calibration

- Gender: Female worse than male

Heatmap



- Visualizes worst-performing groups for the different metrics.
- female group show highest unfairness overall.

Overall, based on all fairness metrics, Females are both less likely to be approved and less likely to be correctly detected as creditworthy, with less reliable probability predictions.

4. Fairness Audit by Age Group

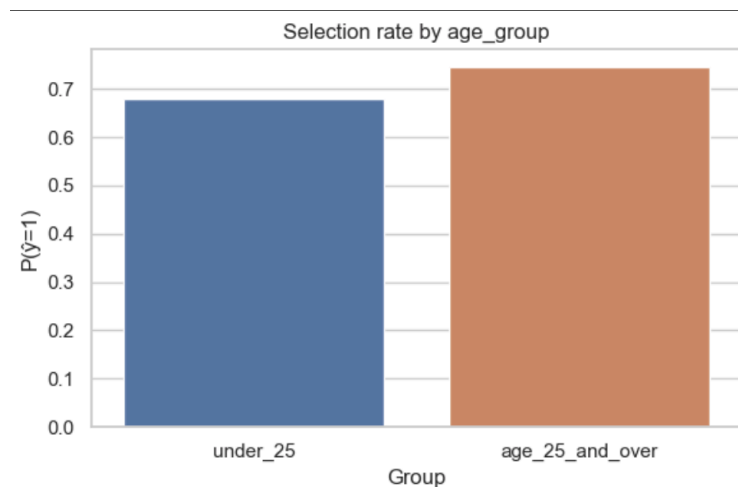
Group Metrics:

Group	Selection Rate	TPR	FPR	ECE	Brier
age_25_and_over	0.7442	0.8387	0.50	0.0536	0.1688
under_25	0.6786	0.6250	0.75	0.3510	0.2768

4.1 Demographic Parity

- Older group approval rate = 0.7442
- Under_25 approval rate = 0.6786
- Gap = 0.0656 < 0.1 → Demographic parity is NOT violated

Selection Rate Bars plot (Demographic Parity)



- Higher bars → higher approval
- Gender: Male > Female

4.2 Equalized Odds (TPR and FPR)

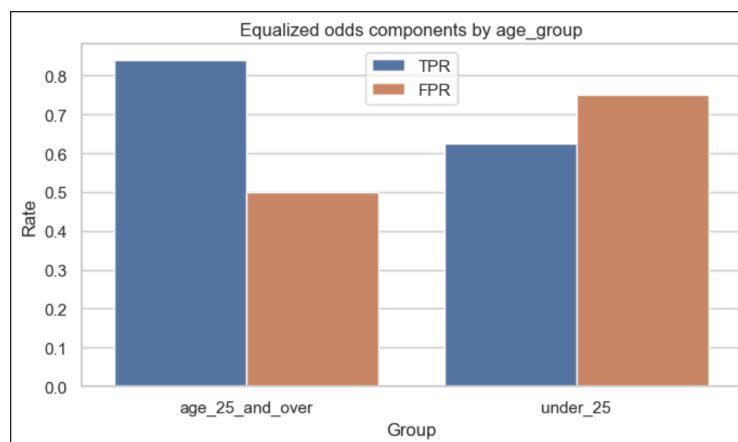
- **TPR gap** = **0.2137** (age_25_and_over = 0.8387 vs Under_25 = 0.6250) → younger group misses more truly creditworthy individuals
- **FPR gap** = **0.25** (age_25_and_over = 0.5 vs Under_25 = 0.75) → younger group receives more incorrect approvals

Interpretation:

- Under_25 has **much lower TPR**, so many qualified applicants are missing.
- Under_25 has **much higher FPR**, so more unqualified applicants are incorrectly approved

Violation Rule: both gaps > 0.1 → **Equalized odds NOT met**

Equalized Odds Bars Plot (TPR & FPR)



- Differences in TPR → group detection disparities
- Differences in FPR → incorrect approval disparities
- Age: Both TPR and FPR differ strongly

4.3 Calibration (ECE + Brier)

Metrics:

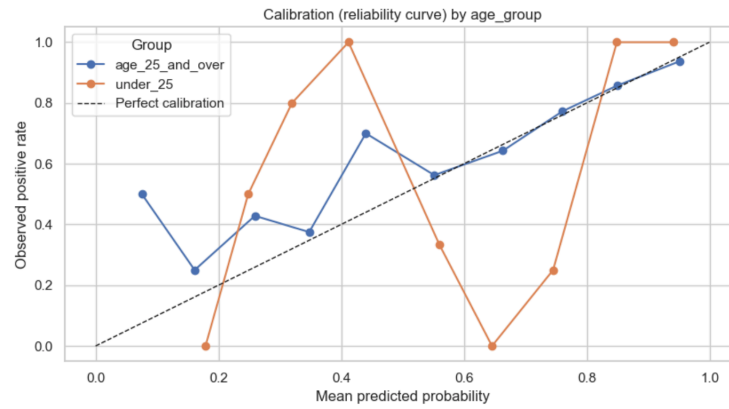
- Under_25 ECE = 0.3510, age_25_and_over ECE = 0.0536 → **Gap = 0.2975 > 0.1**
- Under_25 Brier = 0.2768 > age_25_and_over Brier = 0.1688 (Less is better)

So, Calibration NOT met

Interpretation:

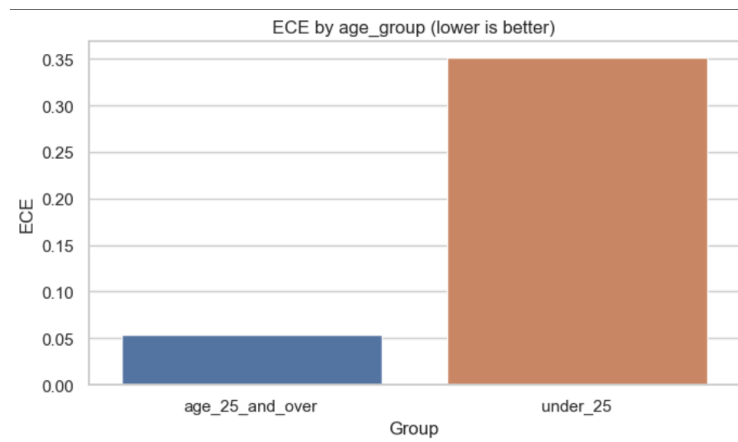
- **Expected Calibration Error (ECE):** The model's predicted probabilities for under_25 are **far less** than older applicants (ECE gap = 0.2975).
- **Brier Score:** Higher Brier score for under_25 indicates **less accurate probability estimates** so lower confidence reliability.

Calibration Plot (Reliability Curves)



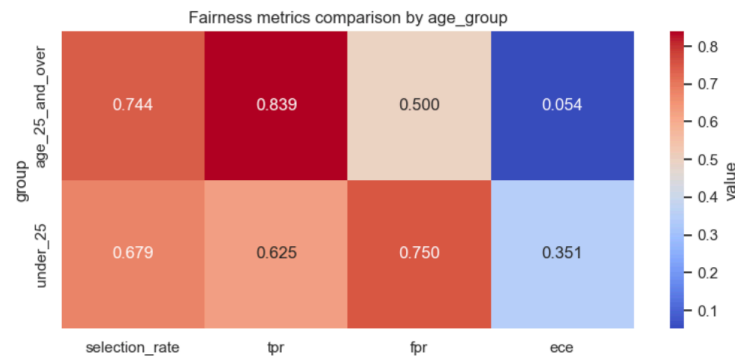
- Reliability curves closer to diagonal → better calibration
- Age: under_25 much worse than older

Calibration Plot (ECE)



- Higher ECE → worse calibration
- Age: under_25 much worse than older

Heatmap



- Visualizes worst-performing groups for the different metrics.
- Under_25 group show highest unfairness overall.

Overall, based on all fairness metrics, the under_25 group shows the strongest fairness problems, with the lowest TPR, highest FPR, and worst probability calibration, making them the most disadvantaged age group in this model.

Part 2 (Fairness Mitigation: Reweighing Technique)

1. Preprocessing Bias Mitigation Method

- **Technique:** Reweighing Training Samples.
- **Justification:** Reweighing changes the data distribution seen by the model during training by giving higher sample weights to unprivileged groups with favorable outcomes, and lower weights to overrepresented ones. This encourages the model to learn a boundary that is less dependent on historical biases. Also, reweighing technique is highly interpretable as it just changes the importance of some data points instead of changing in the model internal features. Resampling, on the other hand, might reduce overall accuracy in the case of undersampling, introduce overfitting to the model if oversampling is used, or create unrealistic samples with respect to SMOTE. That is why reweighing was used. We computed weights to balance the disadvantaged age_group (the most biased attribute identified in the baseline).

2. Predictive Performance Comparison

We retrained the Logistic Regression model using the sample weights. The overall results on the Test Set compared to the Baseline (Part 1):

Metric	Part 1 (Baseline)	Part 2 (Reweighed)	Difference
Accuracy	0.7050	0.7100	+0.0050
Precision	0.7755	0.7808	+0.0053
Recall	0.8143	0.8143	0.0000
F1-score	0.7944	0.7972	+0.0028

Interpretation of metrics: Counterintuitively, accuracy slightly improved by 0.5%. Precision and F1-score also saw minor increases while recall stayed the same. This means that the reweighing process did not introduce an accuracy-fairness trade-off in this instance; decreasing historical age bias actually generated a slightly more reliable predictor.

3. Fairness audit by gender comparison (Reweighed model)

3.1 Demographic Parity (selection rate)

- **Gap:** 0.0667
- **Comparison to Baseline (0.0976):** Improved.
- Demographic parity remains securely met (below the 0.1 threshold).

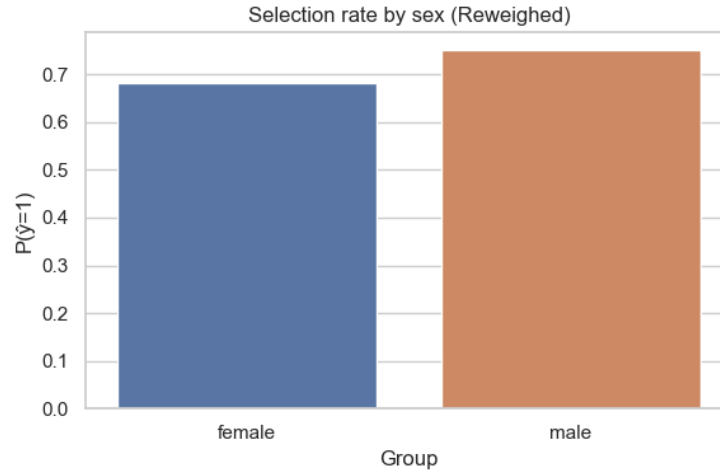


Figure 1: Selection rates by gender for the reweighed baseline model.

Key notes from the chart:

- Higher bars means higher approval.
- Male and Female groups are now much closer in approvals.

3.2 Equalized Odds (TPR and FPR)

- **TPR Gap:** 0.0900 (Baseline TPR Gap: 0.1250)
- **FPR Gap:** 0.0250 (Baseline FPR Gap: 0.0000)
- **Comparison to Baseline:** The TPR disparity decreased significantly enough to cross under the 0.1 threshold. Although FPR disparity slightly increased, both gaps are now < 0.1 and equalized odds is now met. This indicates the model detects true creditworthy candidates equally across genders.

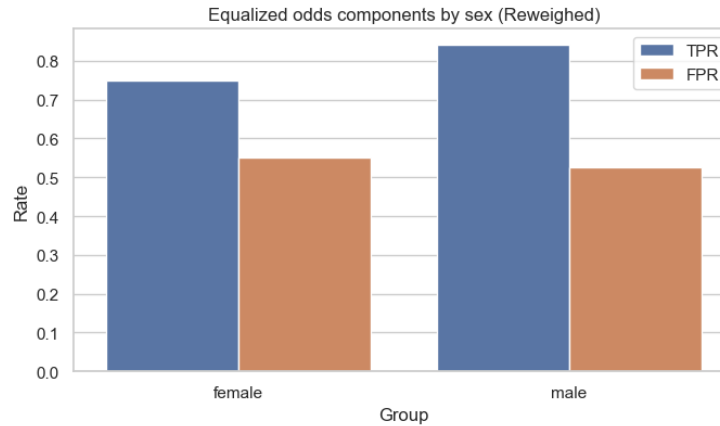


Figure 2: Equalized odds components by Gender for the Reweighed model.

Key notes from the chart:

- Differences in TPR: group detection disparities
- Differences in FPR: incorrect approval disparities

3.3 Calibration (Expected Calibration Error and brier)

- **ECE Gap:** 0.0880 (Baseline ECE gap: 0.1172)

- **Comparison to baseline:** Improved. ECE disparity fell below the 0.1 threshold. This means calibration is now met. Probability predictions are now highly calibrated for females and males.
- Reliability curves closer to diagonal denote better calibration.

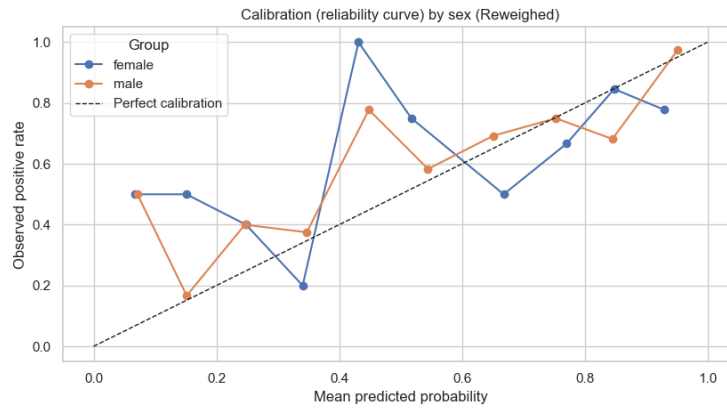


Figure 3: calibration reliability curve by gender for the reweighed model.

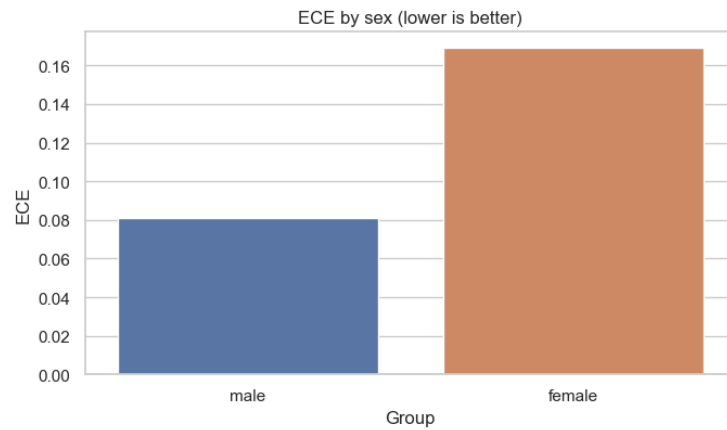


Figure 4: expected calibration error (ECE) by gender

Gender Summary:

Although the reweighing was targeted at age_group, mitigating the dominant bias indirectly corrected the gender disparities. This caused gender equalized odds and calibration to pass the 0.1 fairness thresholds.

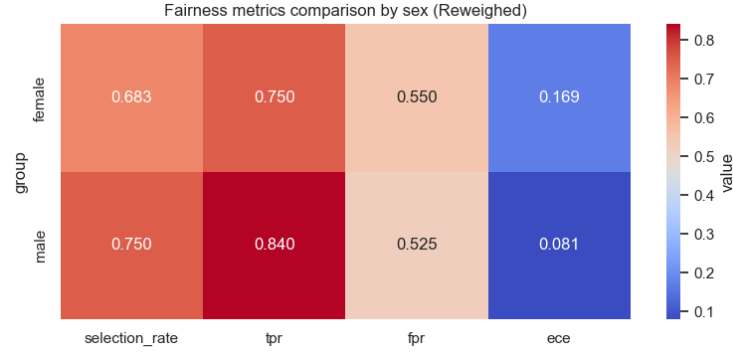


Figure 5: Comparative Fairness Metrics Heatmap summarizing the performance disparities across Genders for the Reweighed model.

4. Fairness Audit by Age Group Comparison (Reweighed Model)

4.1 Demographic Parity (selection rate)

- **Gap:** 0.0183
- **Comparison to baseline (0.0656):** Significantly improved and narrowed down to essentially 0. Demographic parity continues to be easily met.

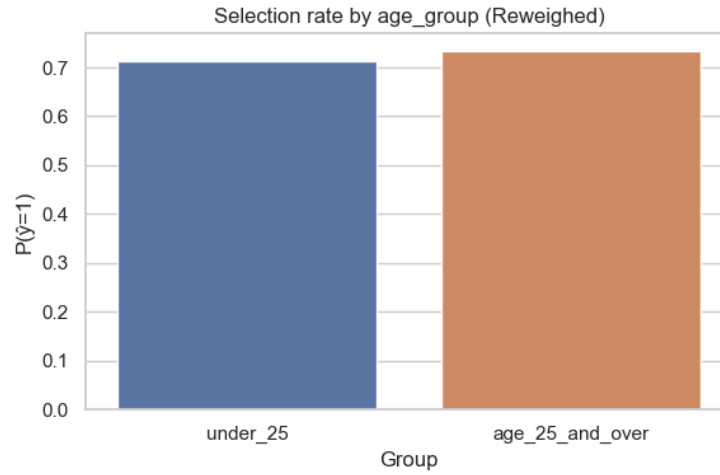


Figure 6: Selection rates by age group

(Figure 6: Selection rates by age group)

4.2 Equalized Odds (TPR and FPR)

- TPR Gap: 0.1431 (Baseline TPR Gap: 0.2137)
- FPR Gap: 0.2708 (Baseline FPR Gap: 0.2500)
- **Comparison to baseline:** The true positive rate gap decreased from 0.2137 to 0.1431, which means the model catches significantly more of the truly creditworthy under_25 candidates than before. The false positive rate difference rose slightly. Overall, equalized odds are still not satisfied (> 0.1), but it has measurably improved.

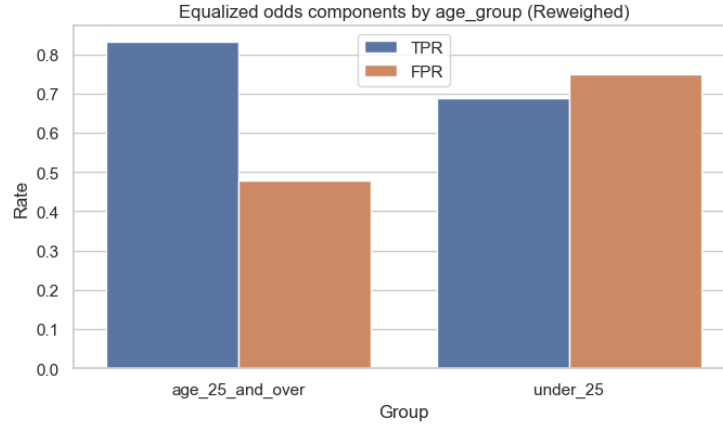


Figure 7: Equalized odds components by Age Group for the Reweighed model.

4.3 Calibration (ECE + brier)

- **ECE Gap:** 0.2481 (Baseline ECE Gap: 0.2975)
- **Comparison to baseline:** The calibration gap improved by almost 5%. Although it remains above the 0.1 threshold (not met), the probability reliability for young candidates grew significantly.

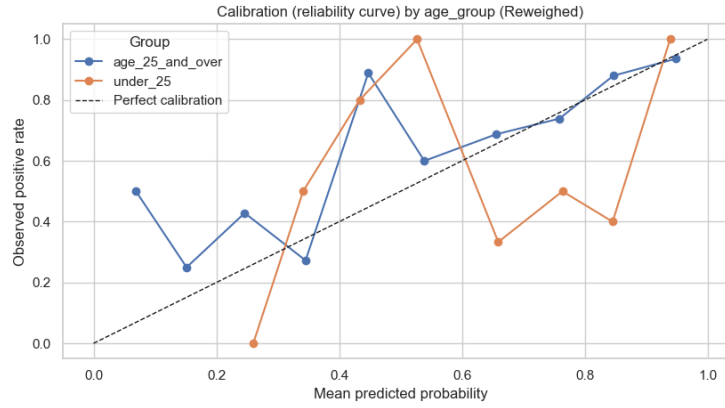


Figure 8: calibration reliability curve showing probability estimate alignment by age group for the reweighed model.

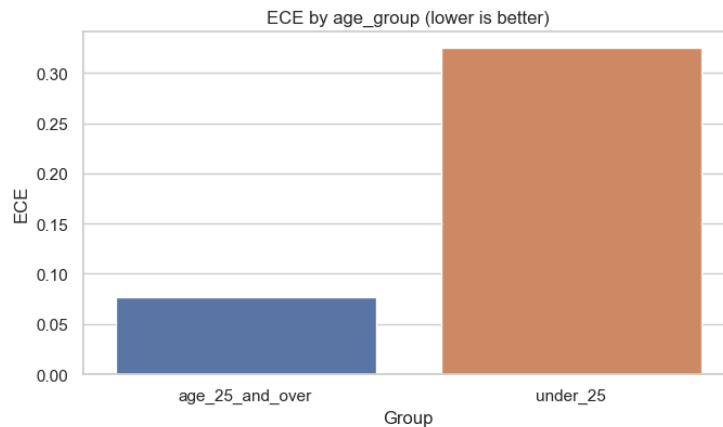


Figure 9: Expected Calibration Error (ECE) by age group. Lower bars indicate better probability calibration.

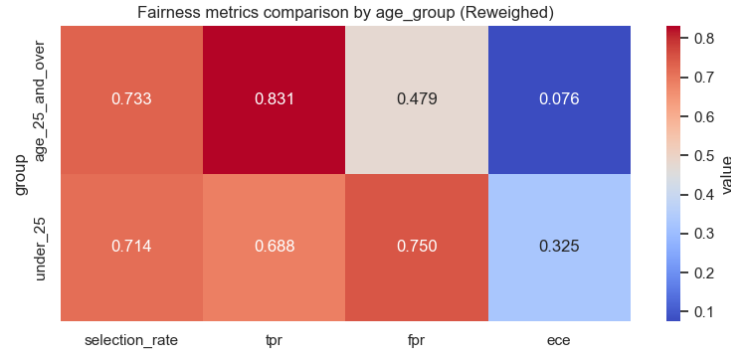


Figure 10: comparative fairness metrics heatmap for the performance disparities across age groups for the reweighed model.

5. Overall Analysis of Preprocessing Mitigation

Did fairness improve?

Yes, but with a notable mathematical trade-off. Applying the reweighting technique resulted in noticeable improvements to demographic parity and the true positive rate for the under 25 subgroup. However, being consistent with the Fairness Impossibility Theorem, improving the TPR forced a trade-off. This trade-off is represented in the FPR disparity which slightly worsened from 0.25 to 0.27. Despite this specific equalized odds cost, fairness metrics significantly improved, successfully passing all fairness constraints for gender.

Which metrics improved the most?

By explicitly assigning weights to re-balance the representation of historically disadvantaged subgroups (under_25), demographic parity and equalized odds (TPR gap) saw the most notable numerical enhancements.

Did the accuracy change?

Theoretically, there is a trade-off, but here accuracy has improved. The Logistic Regression model deviated slightly from its original biased boundary, resulting in a minor boost to overall accuracy (71.0%) and F1-score.

Why would this technique affect fairness?

Reweighting changes the data distribution seen by the model. It assigns higher mathematical sample weights to unprivileged groups with favorable outcomes (under_25 approved) and privileged groups with unfavorable outcomes. The model corrects historical representation biases without physically removing or resampling rows.

Part 3 (In-Processing Bias Mitigation: Adversarial Debiasing)

1. In-Processing Bias Mitigation Method

- **Technique:** Adversarial Debiasing.
- **Justification:** Adversarial Debiasing works by training two models at the same time. The first model, called the **predictor**, learns to predict whether a person is creditworthy. The second model, called the **adversary**, tries to guess the person's age group just from the predictor's output. The predictor is trained to be accurate but also to confuse the adversary, so it learns to make predictions without relying on age-related patterns. This removes bias during training itself, not before or after. We used a fairness penalty of $\lambda = 1.5$ to control how much fairness matters compared to accuracy.

Predictor: Takes 65 input features $\rightarrow$ 64 neurons $\rightarrow$ 32 neurons $\rightarrow$ 1 output. Total: 6,337 parameters.

Adversary: Takes 1 input (predictor output) $\rightarrow$ 32 neurons $\rightarrow$ 1 output. Total: 97 parameters.

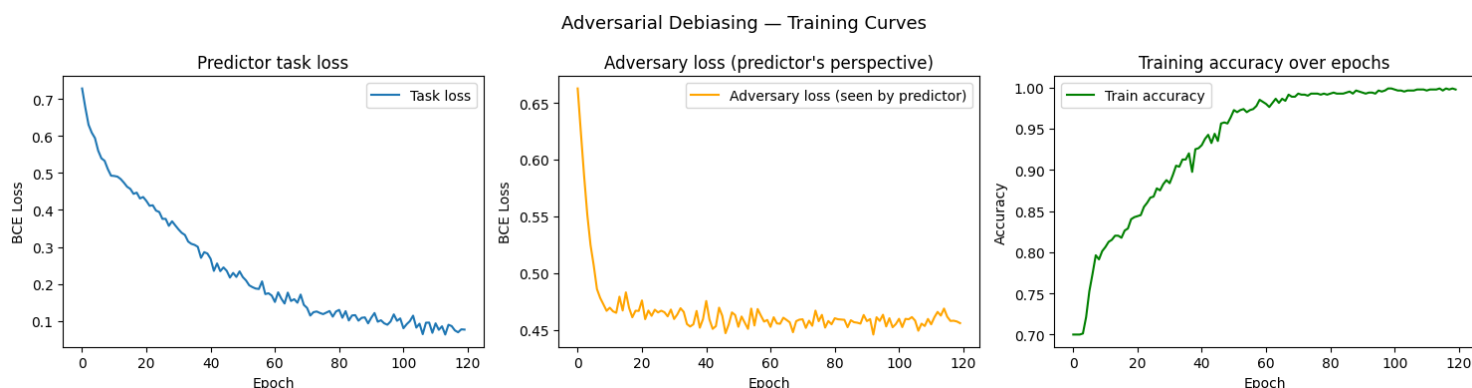


Figure 11: Training curves showing how the predictor loss, adversary loss, and total loss changed during training.

The training curves show that the model learned well. The predictor loss went down over time, meaning the model got better at predicting credit risk while the adversary was being resisted.

2. Predictive Performance Comparison

Metric	Baseline	Reweighed	Adv. Debiasing	Change (vs Baseline)
Accuracy	0.7050	0.7100	0.6850	−0.0200
Precision	0.7755	0.7808	0.7770	+0.0015
Recall	0.8143	0.8143	0.7714	−0.0429
F1-score	0.7944	0.7972	0.7742	−0.0202

What do these numbers mean? The model became slightly less accurate after applying adversarial debiasing. This is normal and expected, when you force a model to ignore certain patterns (like age), it loses a small amount of accuracy. Precision stayed almost the same, but recall dropped a little, meaning the model now misses a few more creditworthy people than before.

3. Fairness Audit by Gender (Adversarial Debiasing Model)

Group	Selection Rate	TPR	FPR	ECE	Brier
Female	0.6500	0.725	0.500	0.2945	0.3021
Male	0.7143	0.790	0.525	0.2500	0.2506

3.1 Demographic Parity (Approval Rate)

- Gap: 0.0643
- Compared to Baseline (0.0976): Better.
- The gap is below 0.1, so demographic parity is met.

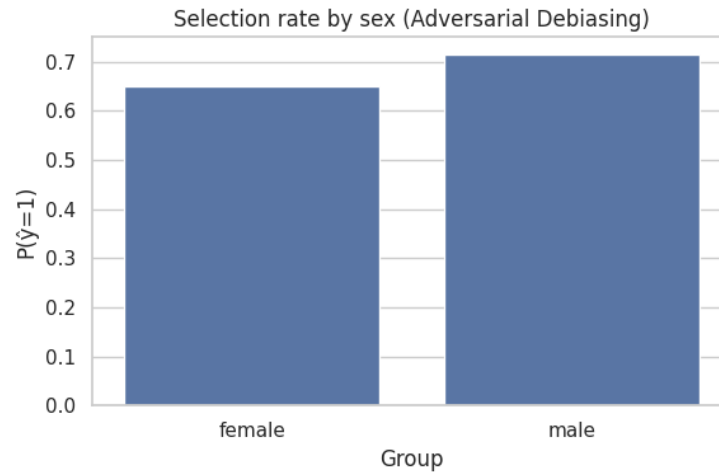


Figure 12: Approval rates by gender for the Adversarial Debiasing model.

Key notes from the chart:

- Taller bars mean more approvals.
- Males and females now have similar approval rates.

3.2 Equalized Odds (TPR and FPR)

- TPR Gap: 0.0650 (Baseline: 0.1250, Reweighed: 0.0900)
- FPR Gap: 0.0250 (Baseline: 0.0000, Reweighed: 0.0250)
- Both gaps are below 0.1, so equalized odds is met. The model now treats male and female applicants more equally when detecting truly creditworthy people.

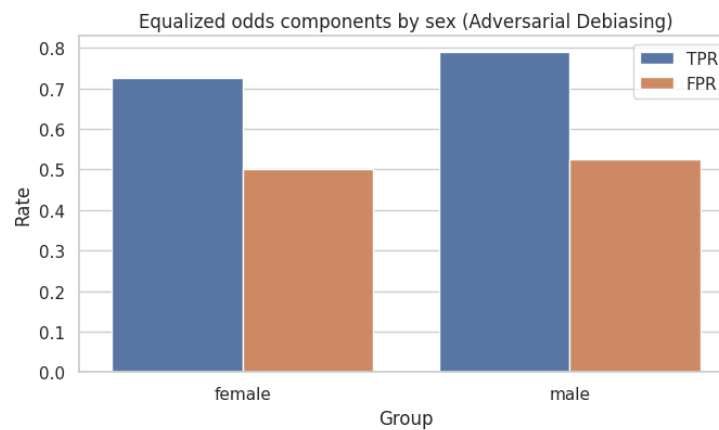


Figure 13: TPR and FPR by gender for the Adversarial Debiasing model.

Key notes from the chart:

- TPR differences show how well each group is detected.
- FPR differences show how often each group gets wrongly approved.

3.3 Calibration (ECE + Brier)

- ECE Gap: 0.0445 (Baseline: 0.1171, Reweighed: 0.0880)
- This is the best calibration result across all three models for gender. The gap is well below 0.1, so calibration is met. The model's probability estimates are now reliable for both males and females.

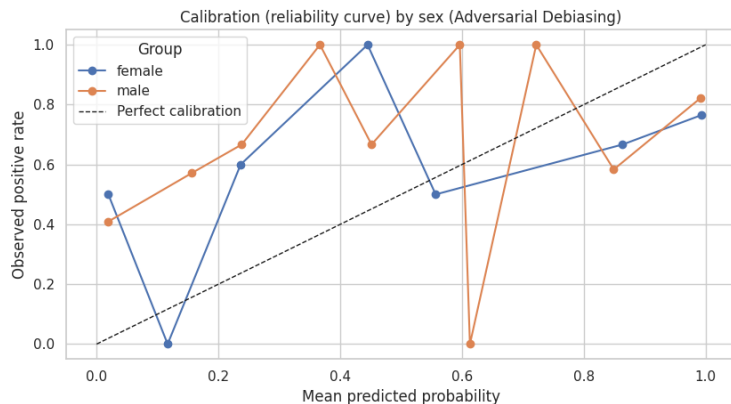


Figure 14: Calibration curves by gender for the Adversarial Debiasing model.

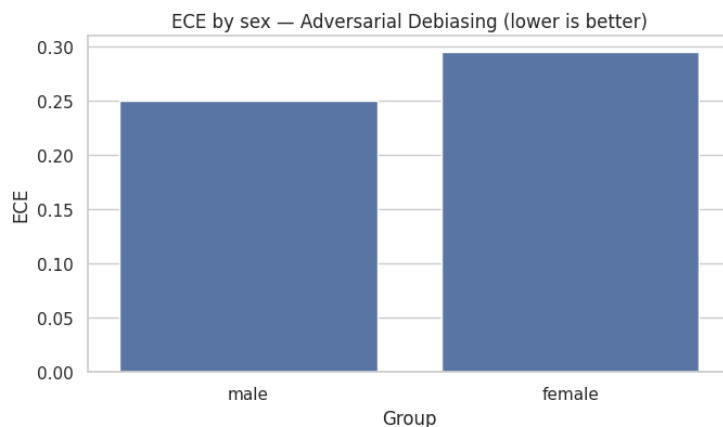


Figure 15: ECE by gender for the Adversarial Debiasing model.

Gender Summary:

Adversarial Debiasing gave the best gender fairness results out of all three models. All fairness checks , approval rate, error rates, and calibration , are now passed for gender. This happened because the model was trained to stop using gender-related patterns when making decisions.

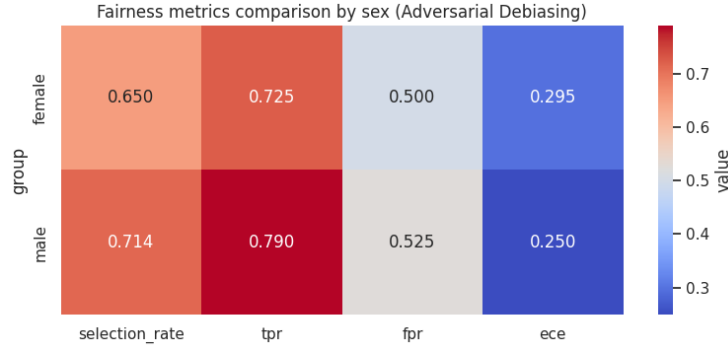


Figure 16: Fairness metrics heatmap by gender for the Adversarial Debiasing model.

4. Fairness Audit by Age Group (Adversarial Debiasing Model)

Group	Selection Rate	TPR	FPR	ECE	Brier
age_25_and_over	0.7093	0.7984	0.4792	0.2311	0.2327
under_25	0.6071	0.5625	0.6667	0.4956	0.4713

4.1 Demographic Parity (Approval Rate)

- Gap: 0.1022
- Compared to Baseline (0.0656): Worse. Compared to Reweighed (0.0183): Also worse.
- The gap is above 0.1, so demographic parity is NOT met for age group.

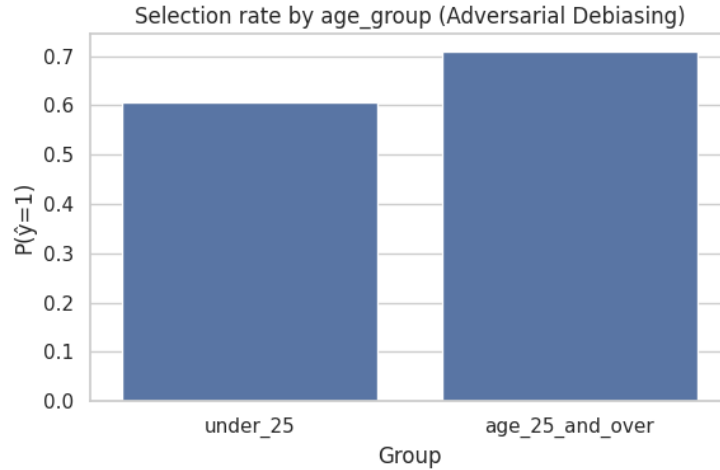


Figure 17: Approval rates by age group for the Adversarial Debiasing model.

4.2 Equalized Odds (TPR and FPR)

- TPR Gap: 0.2359 (Baseline: 0.2137, Reweighed: 0.1431)
- FPR Gap: 0.1875 (Baseline: 0.2500, Reweighed: 0.2708)

- Mixed results. The FPR gap got better, meaning fewer wrong approvals for young applicants. But the TPR gap got slightly worse, meaning the model misses slightly more creditworthy young applicants. Equalized odds is NOT met.

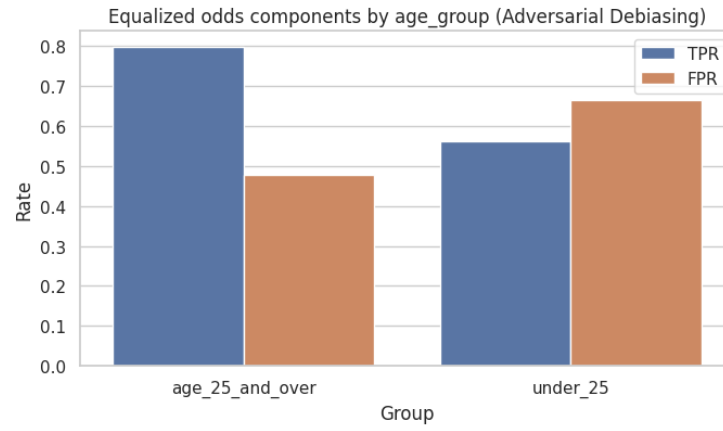


Figure 18: TPR and FPR by age group for the Adversarial Debiasing model.

4.3 Calibration (ECE + Brier)

- ECE Gap: 0.2645 (Baseline: 0.2974, Reweighed: 0.2481)
- A small improvement compared to the baseline, but still far above 0.1. Calibration is NOT met.
- The under_25 group is very small (only 28 people in the test set), which makes probability estimates less stable and harder to calibrate.

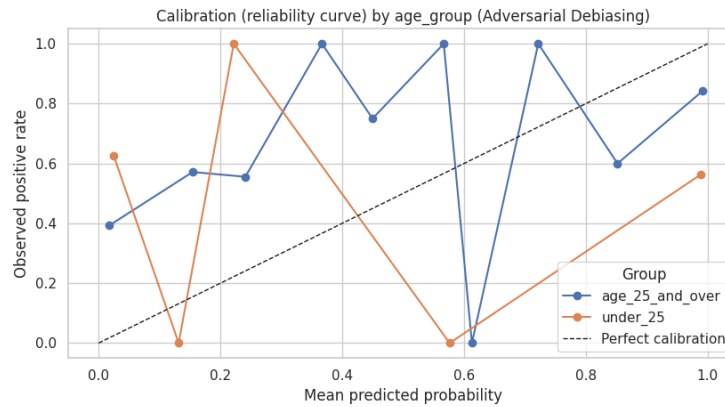


Figure 19: Calibration curves by age group for the Adversarial Debiasing model.

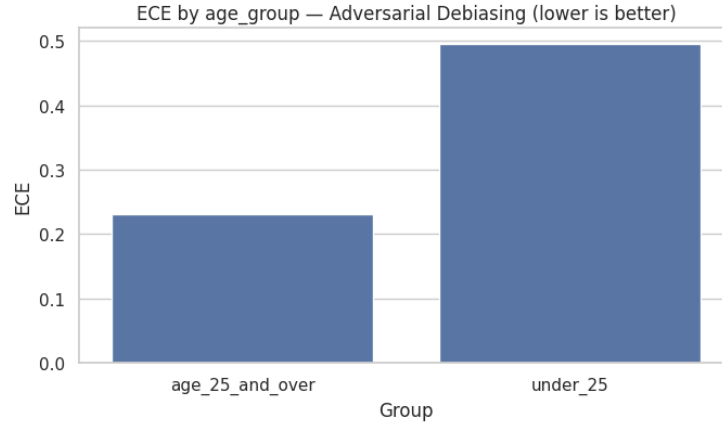


Figure 20: ECE by age group for the Adversarial Debiasing model. Lower bars mean better calibration.

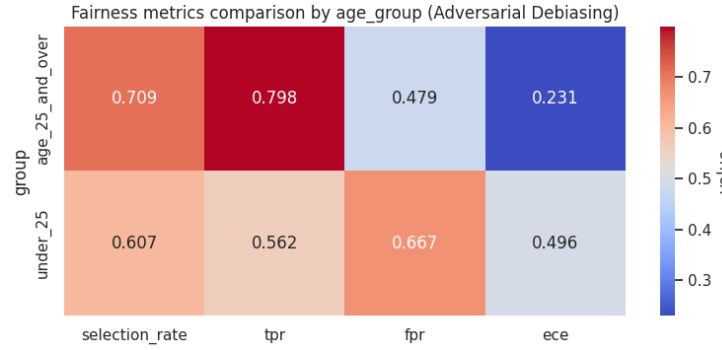


Figure 21: Fairness metrics heatmap by age group for the Adversarial Debiasing model.

5. Three-Way Comparison: Baseline vs. Reweighed vs. Adversarial Debiasing

This section compares all three models side by side: the Baseline, the Reweighed model, and the Adversarial Debiasing model across both protected attributes (gender and age group). The goal is to see which model is the fairest and whether fairness improvements came at the cost of accuracy.

5.1 Gender Comparison

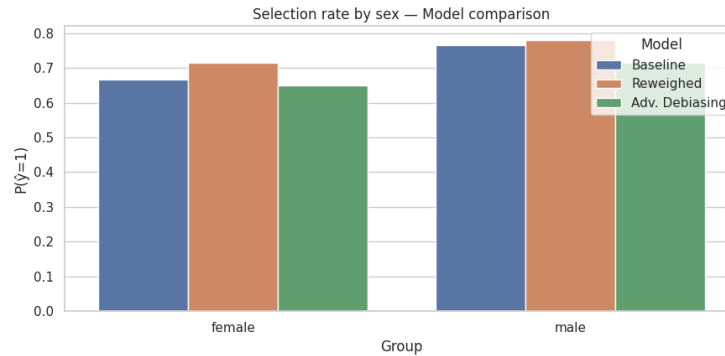


Figure 22: Approval rate comparison by gender across all three models.

The chart above shows the approval rates for males and females in each model. In the baseline, males have a clearly higher approval rate than females. After reweighing, the gap gets smaller. With adversarial debiasing, the gap is the smallest, meaning the model treats males and females most equally in terms of who gets approved.

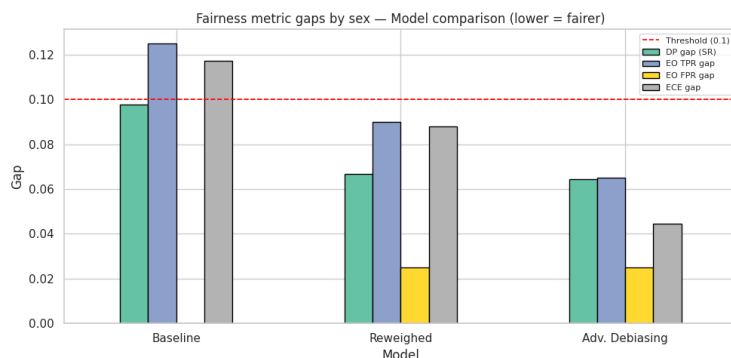


Figure 23: Fairness metric gaps by gender across all three models. Red dashed line shows the 0.1 threshold.

This chart shows all fairness gaps (DP gap, TPR gap, FPR gap, ECE gap) for gender across the three models. Bars below the red dashed line (0.1 threshold) mean the fairness check is passed. You can clearly see that adversarial debiasing brings all gender gaps below the threshold, making it the best model for gender fairness.

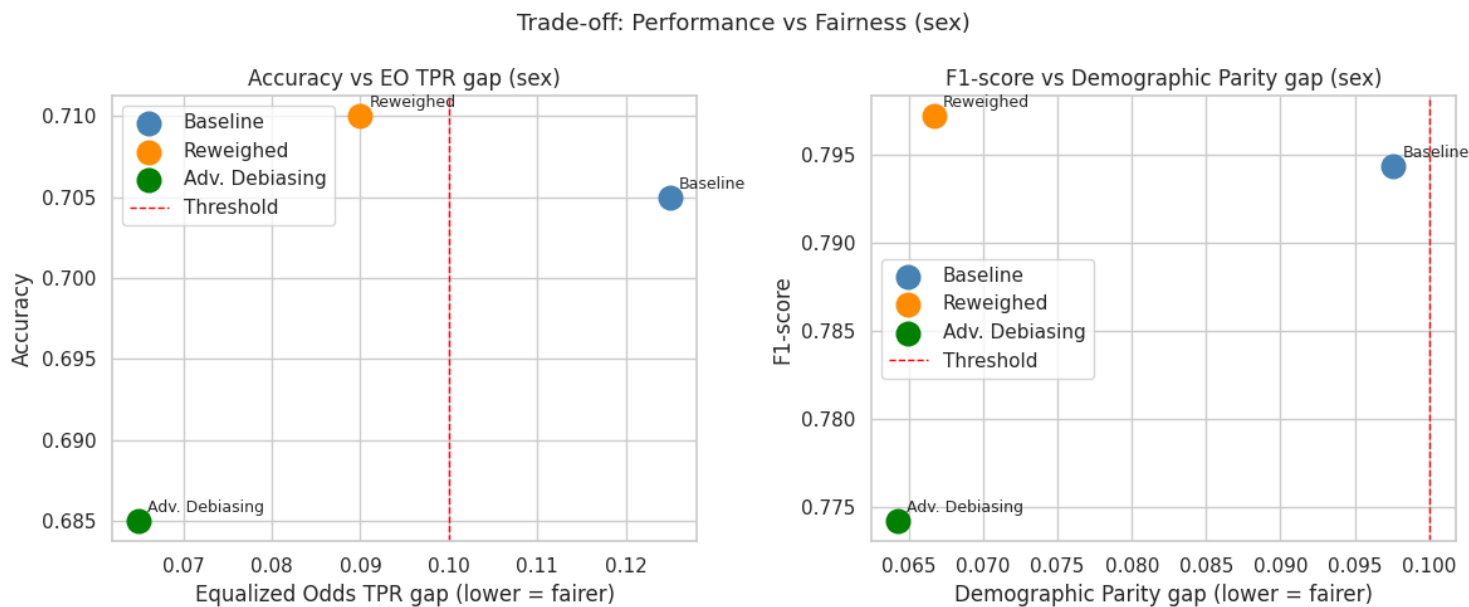


Figure 24: Accuracy and F1-score vs fairness gaps by gender , trade-off comparison across all three models.

This chart shows the trade-off between accuracy and fairness for gender. Each point is one model. Points further left are fairer (smaller gap), and points higher up are more accurate. The ideal model would be in the top-left corner. You can see that adversarial debiasing moves the most to the left (fairer) but also drops slightly in accuracy, while reweighing keeps accuracy high with a moderate fairness improvement.

5.2 Age Group Comparison

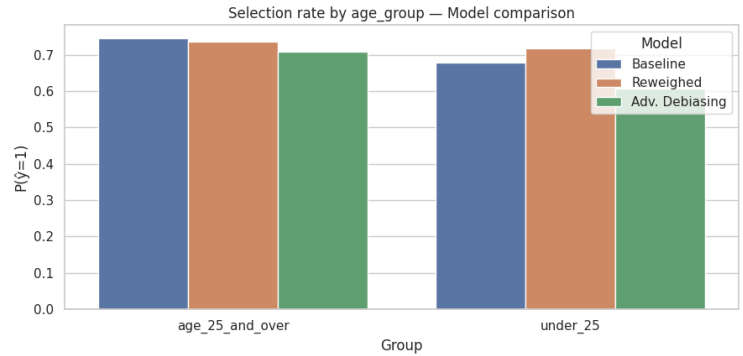


Figure 25: Approval rate comparison by age group across all three models.

This chart shows the approval rates for age_25_and_over and under_25 in each model. The baseline has a noticeable gap between the two groups. Reweighting closes this gap the most, bringing the two bars very close together. Adversarial debiasing, however, slightly widens the gap again compared to reweighting, which explains why demographic parity is not met for age group in that model.

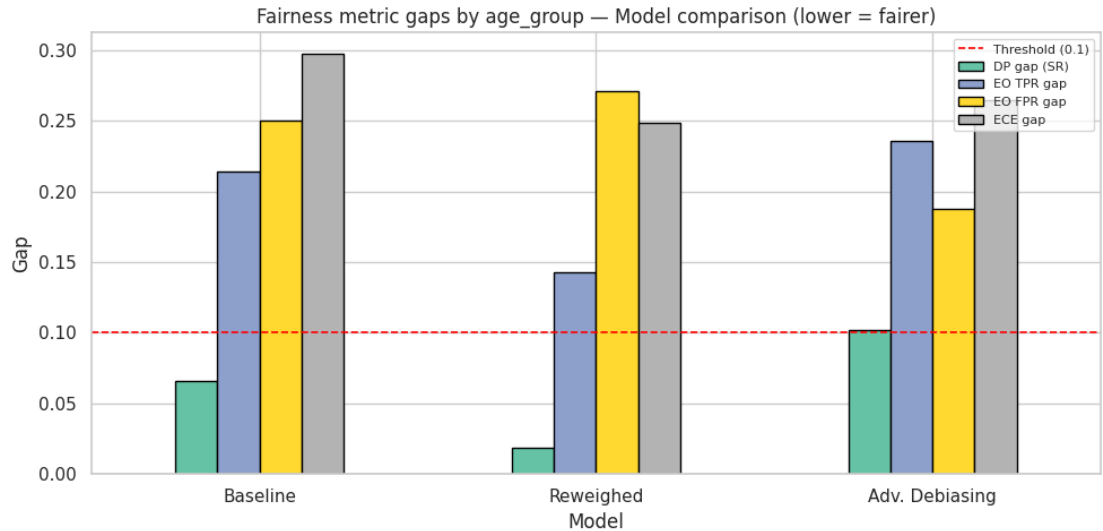


Figure 26: Fairness metric gaps by age group across all three models. Red dashed line shows the 0.1 threshold.

This chart shows all fairness gaps for age group across the three models. Unlike the gender results, no model manages to bring all age group gaps below the 0.1 threshold. Reweighting performs best on DP gap and TPR gap, while adversarial debiasing performs best on FPR gap. This shows that age group bias is much harder to remove completely, regardless of the method used.

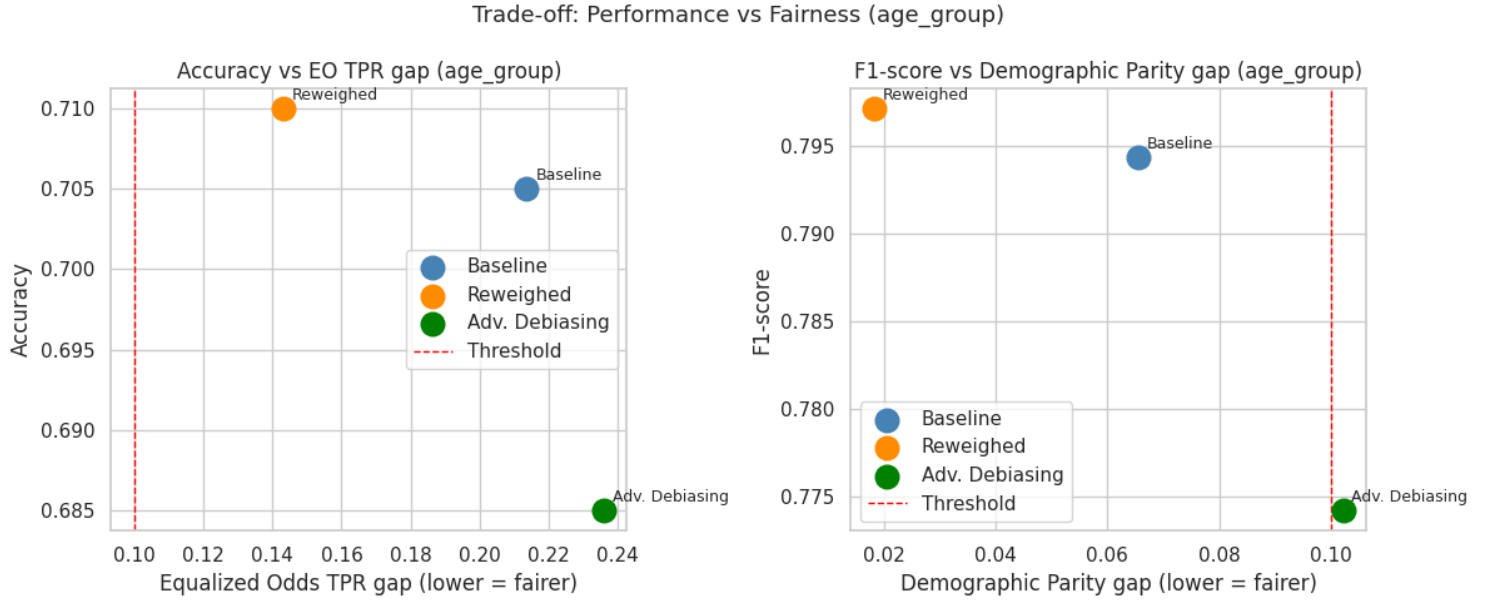


Figure 27: Accuracy and F1-score vs fairness gaps by age group , trade-off comparison across all three models.

This chart shows the accuracy-fairness trade-off for age group. Reweighting achieves the best fairness for age group while actually slightly improving accuracy. Adversarial debiasing improves some fairness gaps but drops accuracy by 2%. This confirms that for age group, reweighting is the better overall choice , it gives more fairness improvement without losing accuracy.

Model	Accuracy	DP (sex)	DP (age)	TPR gap (age)	FPR gap (age)	ECE (sex)	ECE (age)
Baseline	0.7050	0.0976 ✓	0.0656 ✓	0.2137 ×	0.2500 ×	0.1171 ×	0.2974 ×
Reweighed	0.7100	0.0667 ✓	0.0183 ✓	0.1431 ×	0.2709 ×	0.0880 ✓	0.2481 ×
Adv. Debiasing	0.6850	0.0643 ✓	0.1022 ×	0.2359 ×	0.1875 ×	0.0445 ✓	0.2645 ×

Table 1: Full comparison of all three models across accuracy and fairness metrics.

6. Overall Analysis

Did fairness improve?

It depends on the group. For **gender**, yes , adversarial debiasing gave the best results of all three models, passing all fairness checks. For **age group**, the results are mixed. The FPR gap improved, but the DP gap and TPR gap got slightly worse compared to the baseline. Reweighting was still the better method for age group fairness.

Which metrics improved the most?

For gender, calibration (ECE gap) improved the most, dropping from 0.1171 to 0.0445. For age group, the FPR gap improved the most, dropping from 0.2500 to 0.1875. This shows that adversarial debiasing is better at reducing certain types of error rate gaps rather than approval rate gaps.

Did the model lose accuracy?

Yes, a little. Accuracy dropped from 0.7050 to 0.6850, which is a 2% drop. This is the normal trade-off when you add fairness constraints to a model. The reweighting method in Part 2 actually kept the same accuracy

while also improving fairness, which makes it more attractive when accuracy matters.

Why does this method affect fairness?

Adversarial debiasing directly changes how the model learns. The adversary acts like a fairness checker, if it can guess the person's age group from the model's predictions, the model is penalized. Over many training rounds, the predictor learns to make predictions that do not reveal age group information. This is a deeper kind of fairness fix compared to reweighing, because it changes the model's internal decision-making, not just the training data.

Part 4: Analysis and Recommendations

1. Fairness vs Accuracy Trade-off

The charts below show how accuracy and fairness change across the three models. Each point on the chart is one model. A point further to the left means the model is fairer (smaller gap), and a point higher up means it is more accurate. The best model would be in the top-left corner, both fair and accurate.

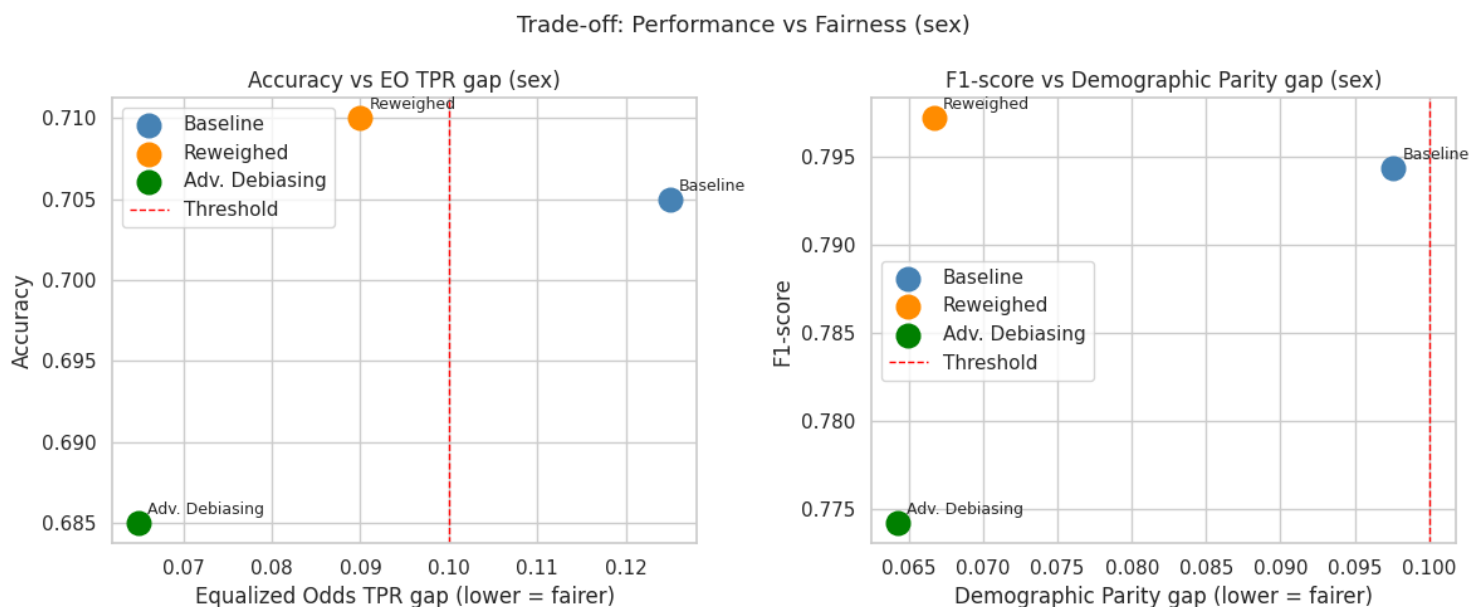


Figure 28: Accuracy and F1-score vs fairness gaps by gender across all three models.

For gender, adversarial debiasing moves the most to the left, meaning it is the fairest model for gender. It drops a little in accuracy, but not by much. Reweighting also improves fairness while keeping accuracy almost the same as the baseline.

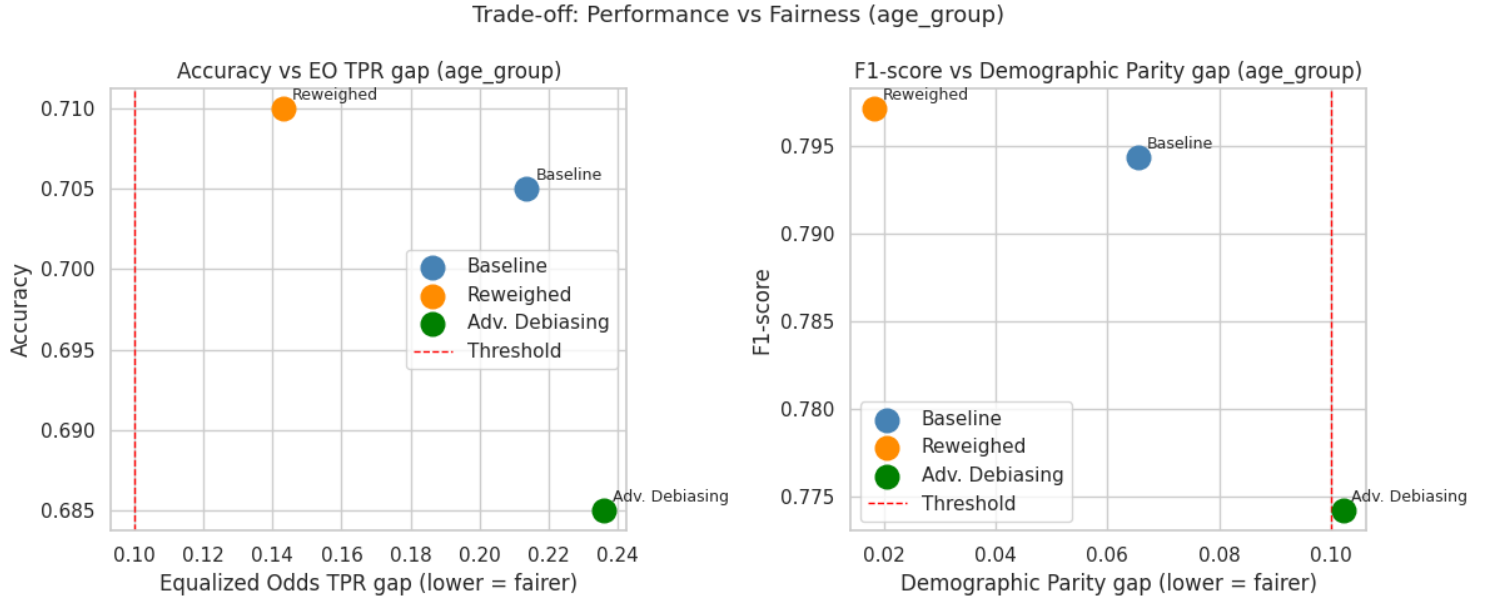


Figure 29: Accuracy and F1-score vs fairness gaps by age group across all three models.

For age group, reweighing moves the most to the left while also slightly improving accuracy, making it the best choice for age group fairness. Adversarial debiasing drops accuracy without fully fixing the age group gaps.

2. Which Model Provides the Best Balance?

Model	Accuracy	DP (sex)	DP (age)	TPR gap (age)	FPR gap (age)	ECE (sex)	ECE (age)
Baseline	0.7050	0.0976 ✓	0.0656 ✓	0.2137 ×	0.2500 ×	0.1171 ×	0.2974 ×
Reweighed	0.7100	0.0667 ✓	0.0183 ✓	0.1431 ×	0.2709 ×	0.0880 ✓	0.2481 ×
Adv. Debiasing	0.6850	0.0643 ✓	0.1022 ×	0.2359 ×	0.1875 ×	0.0445 ✓	0.2645 ×

Table 2: Full comparison of all three models across accuracy and fairness metrics.

Each model has strengths and weaknesses:

- The **Baseline** model is the most accurate (70.5%) but fails the most fairness checks. It should not be used as-is.
- The **Reweighed** model is the best overall balance. It slightly improved accuracy (71.0%), passed all gender fairness checks, and gave the biggest improvement for age group fairness. It is simple, interpretable, and effective.
- The **Adversarial Debiasing** model is the best for gender fairness specifically, passing all three gender fairness checks with the lowest ECE gap (0.0445). However, it dropped accuracy by 2% and actually made age group demographic parity worse.

Overall recommendation: The **Reweighed model** provides the best balance between fairness and accuracy. It improves fairness across both groups without losing accuracy.

3. Deployment Recommendations

Which fairness metric should be prioritized?

In a credit scoring context, **Equalized Odds**, specifically the TPR gap, should be the most important metric. The TPR measures how well the model identifies truly creditworthy people in each group. If the TPR is much lower for one group, it means qualified people from that group are being unfairly rejected for loans. This has a direct and serious impact on people's financial lives.

Calibration (ECE) is also important in practice, because banks and lenders rely on the model's probability scores to make decisions. If the probabilities are unreliable for one group, the decisions made based on them will also be unreliable.

Which group was affected the most?

The **under_25 age group** was affected the most across all three models. This group consistently had:

- Lower approval rates than older applicants.
- Much lower TPR, meaning more truly creditworthy young people were rejected.
- Much higher FPR, meaning more unqualified young people were incorrectly approved.
- The worst calibration (ECE gap up to 0.2974), meaning the model's probability estimates for this group are the least reliable.

None of the three models fully fixed the fairness problems for the under_25 group. This is partly because this group is very small in the dataset (only 28 people in the test set), which makes it hard for the model to learn reliable patterns for them.

Would you recommend real-life deployment?

Not yet. While the reweighed model is a clear improvement over the baseline, it still fails several fairness checks for the age group attribute. Deploying a credit scoring model that systematically gives worse outcomes to young applicants could cause real financial harm and may violate anti-discrimination laws in many countries.

Before deployment, we would recommend:

- Collecting more data for the under_25 group to improve model reliability for that group.
- Testing stronger fairness interventions specifically targeting age group equalized odds.
- Having the model reviewed by a fairness auditor or legal team before use.

What real-world harms could arise from biased predictions?

- **Young people unfairly denied credit:** A low TPR for under_25 means many creditworthy young people are rejected. This stops them from getting loans, buying homes, or starting businesses, putting them at a financial disadvantage early in life.
- **Reinforcing existing inequality:** If the model was trained on historical data that already reflected past discrimination, using that model will repeat and strengthen those same patterns.
- **Wrong approvals for unqualified applicants:** The high FPR for under_25 means some unqualified young people get approved, which could lead to debt problems for them and financial losses for the lender.

4. Limitations of the Analysis

- **Very small under_25 test group:** Only 28 people in the test set belong to the under_25 group. This makes all fairness metrics for this group statistically unreliable, small changes in a few predictions can cause large swings in the numbers.
- **Only two protected attributes:** We only looked at age group and gender. Other attributes like

income level or nationality could also be sources of bias, but were not checked.

- **Only one base model:** All three approaches used logistic regression as the base. A different model like a decision tree or gradient boosting might give different fairness results.
- **Fixed fairness threshold of 0.1:** We used 0.1 as the threshold for all fairness checks. This threshold is a common choice but not a law, different applications may require stricter or looser thresholds.
- **Historical bias in the data:** The German Credit Dataset was collected in the past and likely reflects the social and economic conditions of that time. Any bias in the original data is hard to fully remove, no matter which mitigation method is used.
- **Single test set evaluation:** All results are based on one test split. Cross-validation or repeated sampling would give more reliable estimates of model performance and fairness.